Assignment 2: Simple Search Engine using Hadoop MapReduce

Big Data - IU

This PDF is generated from the verified March 27, 2026 repository run. The submission uses PySpark for preparation, Hadoop Streaming for indexing, Cassandra for storage, and Spark RDD BM25 ranking on YARN.

Methodology

The repository runs a three-service Docker Compose topology: Hadoop/Spark master, one Hadoop worker, and Cassandra. The master entrypoint uploads a parquet slice to HDFS, prepares 1000 plain-text documents in HDFS /data, writes a single-partition /input/data dataset, runs two Hadoop Streaming pipelines, stores the resulting index in Cassandra, and executes three search queries through search.sh on YARN.

The final index is split across four HDFS folders: postings, documents, stats, and vocabulary. Cassandra mirrors these structures with the tables postings, vocabulary, documents, and corpus_stats. The ranker reads the index from Cassandra, parallelizes candidate postings into Spark RDDs, and computes BM25 with k1=1.2 and b=0.75.

Verified Commands

```
docker compose up

docker exec cluster-master bash -lc 'hdfs dfs -ls -R /indexer'
docker exec cluster-master bash -lc 'bash /app/search.sh "dogs"'
docker exec cluster-master bash -lc 'bash /app/search.sh "history of science"'
docker exec cluster-master bash -lc 'bash /app/search.sh "christmas carol"'
```

Observed Counts

```
input_data_line_count
1000

cassandra_counts
postings■251313
vocabulary■40262
documents■1000
corpus_stats■3
```

The verified run completed successfully, produced 1000 prepared input rows, and loaded 251313 postings, 40262 vocabulary terms, 1000 documents, and 3 corpus statistics into Cassandra.

Proof Artifact 1

Verified workflow markers from the successful docker compose up run, including indexing completion, Cassandra counts, query execution, and the final success message.

```
Verified docker compose up workflow

[prepare_data.sh] Found 1000 items
[prepare_data.sh] HDFS /input/data contents:
[create_index.sh] Final HDFS index layout:
{'postings': 251313, 'vocabulary': 40262, 'documents': 1000, 'corpus_stats': 3}
Query: "dogs"
Query: "history of science"
Query: "christmas carol"
[app.sh] Workflow finished successfully. Keeping the master container alive for inspection.

input_data_line_count
1000

cassandra_counts
postings        251313
vocabulary      40262
documents       1000
corpus_stats    3
```

Proof Artifact 2

Verified HDFS index layout produced by create_index.sh after the MapReduce pipelines completed.

Verified HDFS index layout

```
drwxr-xr-x   - root supergroup          0 2026-03-27 14:08 /indexer/documents
-rw-r--r--   1 root supergroup      39936 2026-03-27 14:08 /indexer/documents/part-00000
drwxr-xr-x   - root supergroup          0 2026-03-27 14:08 /indexer/postings
-rw-r--r--   1 root supergroup    6367855 2026-03-27 14:08 /indexer/postings/part-00000
drwxr-xr-x   - root supergroup          0 2026-03-27 14:08 /indexer/stats
-rw-r--r--   1 root supergroup         72 2026-03-27 14:08 /indexer/stats/part-00000
drwxr-xr-x   - root supergroup          0 2026-03-27 14:08 /indexer/vocabulary
-rw-r--r--   1 root supergroup          0 2026-03-27 14:08 /indexer/vocabulary/_SUCCESS
-rw-r--r--   1 root supergroup     649759 2026-03-27 14:08 /indexer/vocabulary/part-00000
```

Proof Artifact 3

Verified YARN-backed search.sh output for the queries dogs and history of science.

```
Verified YARN search results page 1

search.sh "dogs"
26/03/27 14:19:22 INFO ServerInfo: Adding filter to /:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:19:22 INFO ServerInfo: Adding filter to /api:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:19:22 INFO ServerInfo: Adding filter to /jobs/job/kill:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:19:22 INFO ServerInfo: Adding filter to /stages/stage/kill:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:19:23 INFO ServerInfo: Adding filter to /metrics/json:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:19:30 INFO YarnSchedulerBackend$YarnDriverEndpoint: Registered executor
NettyRpcEndpointRef(spark-client://Executor) (172.18.0.2:50180) with ID 1,  ResourceProfileId 0
26/03/27 14:19:30 INFO YarnClientSchedulerBackend: SchedulerBackend is ready for scheduling beginning after
reached minRegisteredResourcesRatio: 0.8
26/03/27 14:19:30 INFO BlockManagerMasterEndpoint: Registering block manager cluster-slave-1:45809 with
229.8 MiB RAM, BlockManagerId(1, cluster-slave-1, 45809, None)
Query: "dogs"
doc_id  title   score
10858097        A_Dangerous_Path        7.897096
35739004        A_Dog's_Purpose 7.781153
6877746 A_Land_Remembered       5.468124
48516501        A_Dog_Named_Gucci       5.399838
7079205 A_Dog_Year      5.378346
17488265        A_Boy_and_His_Dog_(1946_film)   5.200310
6597753 A_II_Z  4.617115
1861460 A_Fine_Romance_(1981_TV_series) 4.557196
372666  A_Confederacy_of_Dunces 3.650317
896285  A_Grand_Day_Out 3.030384

search.sh "history of science"
26/03/27 14:20:11 INFO ServerInfo: Adding filter to /api:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:20:11 INFO ServerInfo: Adding filter to /jobs/job/kill:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:20:11 INFO ServerInfo: Adding filter to /stages/stage/kill:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:20:11 INFO ServerInfo: Adding filter to /metrics/json:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:20:11 INFO YarnSchedulerBackend$YarnSchedulerEndpoint: ApplicationMaster registered as
NettyRpcEndpointRef(spark-client://YarnAM)
26/03/27 14:20:20 INFO YarnSchedulerBackend$YarnDriverEndpoint: Registered executor
NettyRpcEndpointRef(spark-client://Executor) (172.18.0.2:46978) with ID 1,  ResourceProfileId 0
26/03/27 14:20:20 INFO YarnClientSchedulerBackend: SchedulerBackend is ready for scheduling beginning after
reached minRegisteredResourcesRatio: 0.8
26/03/27 14:20:21 INFO BlockManagerMasterEndpoint: Registering block manager cluster-slave-1:45517 with
229.8 MiB RAM, BlockManagerId(1, cluster-slave-1, 45517, None)
Query: "history of science"
doc_id  title   score
69979031        A_History_of_Science,_Technology,_and_Philosophy_in_the_16th_and_17thCenturies  9.308827
60121915        A_Brief_History_of_Everyone_Who_Ever_Lived      8.379427
73922368        A_Honeymoon_in_Space    7.522365
2216485 A_History_of_Knowledge  7.521874
69751062        A_Fortune_for_Kregen    7.501041
40541172        A_History_of_All_Nations_from_the_Earliest_Times        6.757579
67078438        A_History_of_Nomadic_Behavior   5.995210
8191051 A_History_of_the_Life_and_Voyages_of_Christopher_Columbus       5.814787
32151667        A_Gnome_There_Was       5.718227
21141177        A_Key_into_the_Language_of_America      5.683835
```

Proof Artifact 4

Verified YARN-backed search.sh output for the query christmas carol.

```
Verified YARN search results page 2

search.sh "christmas carol"
26/03/27 14:21:03 INFO ServerInfo: Adding filter to /api:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:21:03 INFO ServerInfo: Adding filter to /jobs/job/kill:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:21:03 INFO ServerInfo: Adding filter to /stages/stage/kill:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:21:03 INFO ServerInfo: Adding filter to /metrics/json:
org.apache.hadoop.yarn.server.webproxy.amfilter.AmIpFilter
26/03/27 14:21:03 INFO YarnSchedulerBackend$YarnSchedulerEndpoint: ApplicationMaster registered as
NettyRpcEndpointRef(spark-client://YarnAM)
26/03/27 14:21:14 INFO YarnSchedulerBackend$YarnDriverEndpoint: Registered executor
NettyRpcEndpointRef(spark-client://Executor) (172.18.0.2:37086) with ID 1,  ResourceProfileId 0
26/03/27 14:21:14 INFO YarnClientSchedulerBackend: SchedulerBackend is ready for scheduling beginning after
reached minRegisteredResourcesRatio: 0.8
26/03/27 14:21:14 INFO BlockManagerMasterEndpoint: Registering block manager cluster-slave-1:42543 with
229.8 MiB RAM, BlockManagerId(1, cluster-slave-1, 42543, None)
Query: "christmas carol"
doc_id  title   score
12212399        A_Flintstones_Christmas_Carol   13.382449
16456979        A_Christmas_Carol_(1982_film)   13.358937
7868751 A_Christmas_Carol_(2006_film)   13.301650
73670   A_Christmas_Carol       13.224560
36534132        A_Christmas_Carol_(Shower_of_Stars)     12.811424
8468715 A_Christmas_Carol_(1997_film)   12.615798
25923604        A_Child's_Christmases_in_Wales  10.272921
29045544        A_Christmas_Cornucopia  10.203559
65714859        A_Drummer_Boy_Christmas 9.886462
20357995        A_Letter_from_Home_(film)       6.911495
```

Discussion

The retrieved results are qualitatively sensible: dog-related titles dominate the dogs query, history-themed titles dominate history of science, and Christmas Carol variants dominate christmas carol. The implementation therefore satisfies both the structural assignment requirements and the expected end-to-end behavior.